

Linear algebra: Exercise session 5

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1. Are the following matrices invertible? If so, what are their inverses?

$$A = \begin{bmatrix} -3 & 5 & 1 \\ 1 & 6 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & 1 & 4 \\ 0 & 1 & 0 \\ 4 & 9 & 4 \end{bmatrix}, \quad D = \begin{bmatrix} 4 & 1 & 2 \\ 2 & 4 & 1 \\ 8 & -7 & 4 \end{bmatrix}, \quad E = \begin{bmatrix} -1 & 5 & 2 \\ 0 & 6 & -1 \\ 1 & 1 & -3 \end{bmatrix},$$
$$F = \begin{bmatrix} -4 & 1 & -2 \\ 2 & -4 & -1 \\ 1 & -2 & -4 \end{bmatrix}, \quad G = \begin{bmatrix} -3 & 1 & -3 \\ 1 & -1 & 2 \\ 2 & 1 & 0 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & -1 & -8 & 0 \\ 0 & 4 & -1 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 7 \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 7 \end{bmatrix}.$$

2. Solve the following systems of linear equations by using the inverse matrix.

$$(a) \begin{cases} 3x_1 + 8x_2 = 5, \\ 4x_1 + 11x_2 = 7, \end{cases} \quad (b) \begin{cases} 5x_1 + 15x_2 + 56x_3 = 35, \\ -4x_1 - 11x_2 - 41x_3 = -26, \\ -x_1 - 3x_2 - 11x_3 = -7. \end{cases}$$

3. Prove the following statements:

- (a) The $n \times n$ matrix A is invertible (nonsingular) if and only if $\text{rank}(A) = n$ if and only if $N(A) = \{0\}$ ($\dim(N(A)) = 0$).
 - (b) If A is an $n \times n$ invertible matrix such that $A^2 = A$, then it is the identity matrix.
 - (c) If A is an $n \times n$ matrix with $A^n = 0$, for some number n , then it is noninvertible.
 - (d) If A is an $n \times n$ matrix with $A^n = 0$, for some number n , then the matrix $(I - A)$ is invertible and its inverse is $(I + A + A^2 + \cdots + A^{n-1})$.
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4. Let A, B and X be three square invertible matrices of the same size. Find an expression of X in the following cases:

- (a) $5A - X = I - 5X$.
 - (b) $2AX^{-1} + B = 0$.
 - (c) $AXB = A + B$.
 - (d) $AX^{-1} + X^{-1} = B$.
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5. Consider the following matrix M , where A is a $q \times p$ and B is an $r \times q$ matrix. What is the inverse of M and N ?

$$M = \begin{bmatrix} I_p & 0 \\ A & I_q \end{bmatrix}, \quad N = \begin{bmatrix} I_p & 0 & 0 \\ A & I_q & 0 \\ 0 & B & I_r \end{bmatrix}.$$

6. Compute the inverse for the following matrices

$$L = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ c & b & 1 \end{bmatrix}, \quad D = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & a' & c' \\ 0 & 1 & b' \\ 0 & 0 & 1 \end{bmatrix},$$

where $d_1, d_2, d_3 \neq 0$.

Is there an easy way to compute the inverse of any matrix A which can be written as $A = LDU$?